Notes

Cloud computing

- Cloud computing has become essential, with most major enterprises use it frequently to lower their capital costs and day-to-day expenses, while enabling online powerful applications.
- Organizations often use a combination of public cloud, and private cloud solutions in what is termed hybrid cloud, and commonly have more than one cloud provider, which is known as multicloud.
- With so many types of cloud computing in use simultaneously across an organization, there is quite a bit of complexity, and the potential for these clouds to turn into a storm, with data as well as applications in use across multiple cloud solutions.

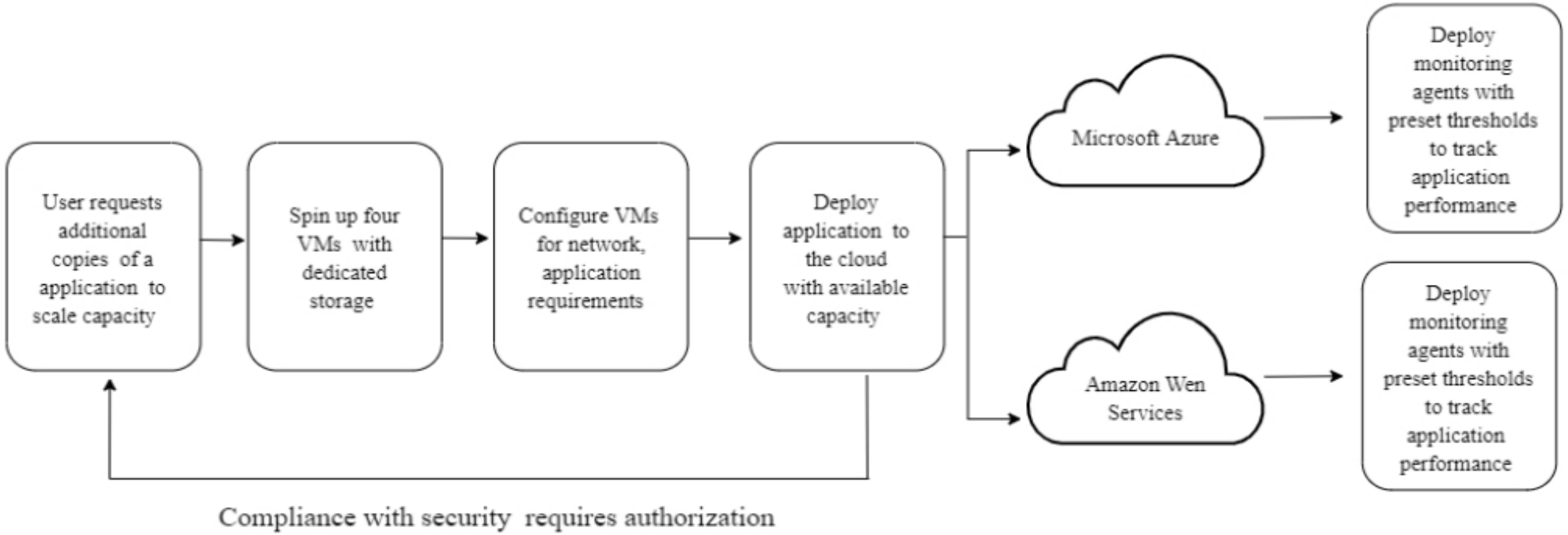
Cloud orchestration

- Cloud orchestration is the process to manage these multiple workloads, in an automated fashion, across several cloud solutions with the goal being to synthesize this into a single workflow.
- Just like the challenge of an orchestra is to get the all instruments to perform the same piece together, this makes cloud orchestration like the musical conductor that controls the performance and keeps it in sync.
- This can be quite challenging as cloud orchestration involves heterogeneous systems, across both public and private cloud solutions, which as they are often geographically separated requires managing the network that connects them as well.

Cloud orchestration

What is Cloud Orchestration

Cloud orchestration enforces the workflow order to automate tasks and enhances security with identity and access management policies. It can also unite disparate cloud deployment to work together for a given workload.



Cloud orchestration vs. Cloud automation

- Cloud automation concerns the control a single task. This requires manual coding to automate the process, which is a standalone process and a single workload.
- Cloud orchestration, on the other hand, involves the coordination of multiple tasks. The manual coding requirements are frequently less, as the coding from cloud automation is built upon, which avoids redundant work.
- Compared to cloud automation, cloud orchestration works on a higher level of coordination as the processes are already streamlined through cloud automation.
- When both used together well, workloads are supported at a high efficiency, and at a reduced cost.

Cloud orchestration models

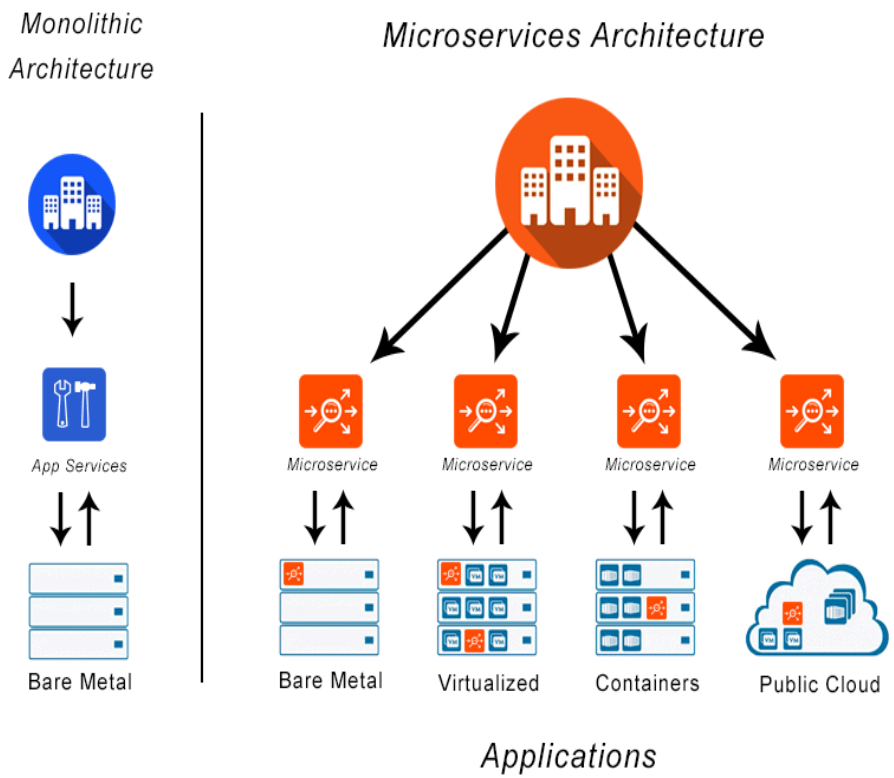
- There are two types of models for cloud orchestration: single cloud, and multicloud.
- For the **single cloud model**, multiple applications all get run on the same cloud service provider, which is a simpler setup.
- The more complicated, but also more powerful model is the **multicloud setup**. Here we have multiple applications, which are located on different cloud platforms, and multicloud orchestration interconnects them so they can perform as a single system, with the advantage of high redundancy.

Cloud orchestration benefits

- **Cost reductions -** By coordinating the processes, it reduces the amount of IT resources required, and the number of system administrators.
- **Capacity increase** - Physical resources are coordinated with virtual resources, so that overall capacity is increased for times of peak demand for rapid scaling just when it is needed.
- **Efficiency** - Cloud orchestration simplifies management of repetitive tasks through automation. This generally requires less IT staff, so they can be reassigned for other tasks.
- **Better connections** - Cloud orchestration provides interconnections to heterogeneous devices.

Microservices

Microservices is an architectural design for building a distributed application using containers. They get their name because each function of the application operates as an independent service. This architecture allows for each service to scale or update without disrupting other services in the application.

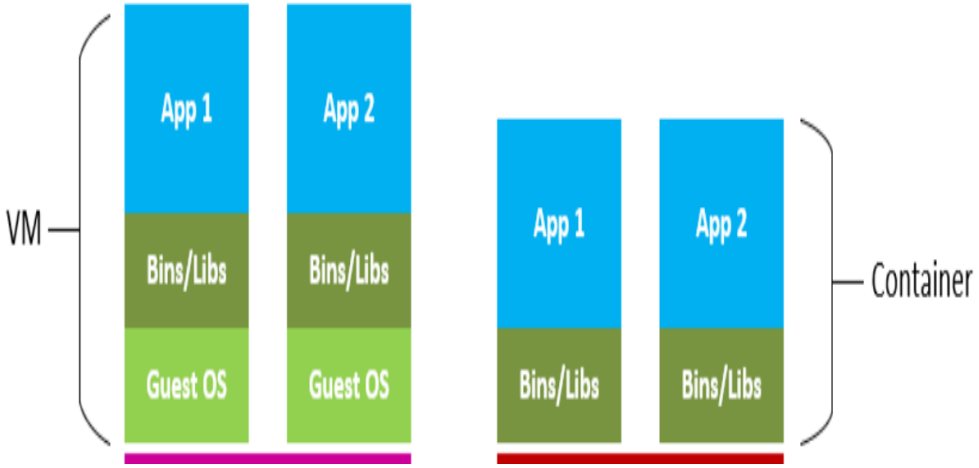


A bare-metal server is a physical computer server that is used by one consumer, or tenant, only.

Containers

- Containers are packages of software that contain all of the necessary elements to run in any environment. In this way, containers virtualize the operating system and run anywhere, from a private data center to the public cloud or even on a developer's personal laptop.

- Containers are a form of operating system virtualization. A single container might be used to run anything from a small microservice or software process to a larger application. Inside a container are all the necessary executables, binary code, libraries, and configuration files.



Container removes the need to run a separate operating system for each application, allowing for higher resource utilization and lower costs.

Application agility

- Agile applications are deconstructed, distributed and dynamically assembled as needed.
- The whole point of microservices and containers is also to re-move dependencies on specific servers and hardware components.

Deployment Options

On-Premises Infrastructure

- This is been the dominant enterprise computing model for more than 50 years.
- Organizations maintain their own equipment and software in a captive data center with full control over all aspects of processing, scheduling, administration and maintenance.
- Many organizations in regulated industries have no choice but to use an on-premises model be-cause of the need to tightly control data and document processes.

Public Cloud

- Public cloud makes resources, such as processors, memory, operating systems, applica-tions and storage, available over the public internet on a pay-per-usage basis. Think of it asa computers in the sky.
- Public cloud is like using a local server, but the server is virtualized and managed elsewhere by a cloud provider with a high degree of automation.
- Public cloud instances can be launched with afew mouse clicks and just as easily taken down when no longer needed.

Public Cloud

- Billing is usually based upon usage, which gives organiza-tions accountability and flexibility to pay only for the resources they use.
- Public cloud in-stances can be scaled up or down with relative ease.

Private Cloud

- For organizations that want the flexible automation benefits of public cloud but need to keep resources on premises for control or compliance reasons, private cloud is a popular alternative.
- This model provides the same scalability, automation and flexibility advantages of public cloud and on-premise environment that can be physically secured and tightly managed.
- Private clouds can be built using existing data center equipment architecture or licensed from public cloud providers, could deliver what is essentially a version of their existing services in a secure environment.

Hybrid Cloud

- When you combine a public and private cloud, you get a hybrid cloud. This architecture combines both models in a manner that is seamless and that permits workloads to easily move back and forth.

- This gives organizations a combination of control and flexibility that can be adjusted to the situation. Hybrid architecture preserves existing hardware and software investments while giving companies the flexibility to move applications to the cloud as resources and budgets permit.

Containers and Clouds

- Containers are an ideal tool for developers to use when shifting between on premises, private cloud and public cloud architectures.
- Because containers are independent of the underlying operating system and infrastructure, they can be moved quickly and with minimal disruption.
- Some organizations even use multiple public clouds, an shift workloads back and forth depending upon price and special offers from the service providers.
- Containers make this process simple

Orchestration

- Agile infrastructure should minimize the need for human intervention in routine tasks, such as resource deployment and management. Overworked IT administrators and paper-work can introduce significant delay that undermines the value of cloud environments. Automation makes cloud computing fast and efficient by using software tools to handle these tasks.

- For example, automation can enable the setup of multiple virtual machines with identical configurations using a single script written in Puppet, an open-source configuration tool that enables applications and infrastructure to be defined using English-like com-mands. Puppet scripts can be shared and used to enforce changes across data center and cloud platforms.

Orchestration

- Ansible is a radically simple IT automation platform that makes your applications and systems easier to deploy and maintain. Automate everything from code deployment to network configuration to cloud management. It can also be used to automate cloud provisioning and intra-service orchestration.
- Kubernetes is bringing these same kinds of automation and orchestration capabilities to containers. Kubernetes isn’t a replacement for Puppet and Ansible, but is another resource that works specifically at the container layer and that can be managed by those automation tools.
- The combination of VM automation and Kubernetes gives IT organizations unprecedented productivity advantages compared to manual systems administration.

Edge Computing

- The internet of things will create vast new volumes of data, a flood that International Data Corp. expects will reach 44 zettabytes annually by 2020.
- These unprecedented data volumes will require a new approach to processing, since traditional server, storage and network models won’t scale enough. This is why edge computing is rapidly emerging as a new architecture.
- Edge computing is a form of computing that is done on site or near a particular data source, minimizing the need for data to be processed in a remote data center.

Edge Computing

- Edge computing distributes resources to the far reaches of the network and close to the devices that generate data. Edge servers collect streaming data, analyze it and make decisions as necessary. These servers can pass selected or summary data to the cloud over the network, but most of the processing takes place locally.

Serverless computing

- Serverless is a cloud-native development model that allows developers to build and run applications without having to manage servers.
- There are still servers in serverless, but they are abstracted away from app development. A cloud provider handles the routine work of provisioning, maintaining, and scaling the server infrastructure. Developers can simply package their code in containers for deployment.
- Serverless computing is a method of providing backend services on an as-used basis. A serverless provider allows users to write and deploy code without the hassle of worrying about the underlying infrastructure. A company that gets backend services from a serverless vendor is charged based on their computation and do not have to reserve and pay for a fixed amount of bandwidth or number of servers, as the service is auto-scaling. Amazon Web Services' Lambda is an example of serverless computing